



AGRICULTURAL PRICE INTELLIGENCE PLATFORM

Smart Mandi Selection Platform

Official User Manual & Operating Guide

Real-Time Price, Transport & Cost Intelligence for Farmers

AUTHOR & LEAD ARCHITECT

Devansh Rahatal

Live Application: smart-mandi-selection.vercel.app

API Documentation: smart-mandi-selection.onrender.com/docs

Tech Stack: React 18 (Vite) · FastAPI · Leaflet GIS · SQLAlchemy · Twilio WhatsApp · Voice AI (gTTS)

Document Version 1.0 · Production Release

Table of Contents

- 1 The Problem: “The Price Illusion” in Indian Agriculture3**
 - 1.1 The Critical Flaws of the Traditional System.....3
 - 1.2 Real-World Proof: Jaipur Tomato Farmer (20 Quintals)3
- 2 The Solution Engineered by Devansh Rahatal4**
 - 2.1 How the Engine Thinks: From Raw Price to True Profit.....4
 - 2.2 Core Technological Innovations4
- 3 Step-by-Step User Walkthrough5**
 - 3.1 Web Application Walkthrough5
 - 3.2 WhatsApp Voice & Text AI Assistant Walkthrough7
- 4 Technical Setup & Local Run Guide.....8**
 - 4.1 Prerequisites.....8
 - 4.2 Run Backend Locally8
 - 4.3 Run Frontend Locally8

1 The Problem: “The Price Illusion” in Indian Agriculture

In Indian agriculture, over 86% of farmers are small and marginal producers. When deciding where to sell their harvest, farmers rely on traditional government portals or word-of-mouth price lists that display only the Gross Modal Price — the raw advertised auction price at a mandi, with none of the real costs of getting there.

1.1 The Critical Flaws of the Traditional System

1. **The Price Illusion Trap** — An advertised price of ₹2,900/quintal in a distant major metro mandi looks vastly superior to ₹2,450/quintal in a nearby district mandi. Chasing this illusion, farmers hire dedicated transport and travel hundreds of kilometers.
2. **Unaccounted Logistics & Hidden APMC Deductions** — Transport freight runs ₹12–₹18/km and multiplies rapidly with distance; APMC market cess and middleman commissions vary from 1.5% to 8% across states; and unregulated loading, unloading, weighing, and bag-stitching levies are charged per quintal.
3. **Perishable Crop Spoilage in Transit** — Highly perishable commodities such as tomatoes, bananas, and onions degrade with every hour spent in transit under heat and bumpy road conditions, causing direct weight loss and quality downgrades.

1.2 Real-World Proof: Jaipur Tomato Farmer (20 Quintals)

The table below compares three candidate APMC markets for the same 20-quintal tomato harvest, showing how the highest advertised price produces the lowest real return.

Candidate APMC Market	Advertised Price	Transport	Commission	Spoilage	Net Profit / q	Verdict
Kota Krishi Mandi (Nearby)	₹2,450	-₹484	-₹110	-₹79	₹1,737.16	OPTIMAL
Azadpur, Delhi (Distant Hub)	₹2,901	-₹927	-₹218	-₹119	₹1,568.06	Price Trap
Vashi, Mumbai (Metro Hub)	₹2,820	-₹2,347	-₹197	-₹420	₹0.00	Net Loss

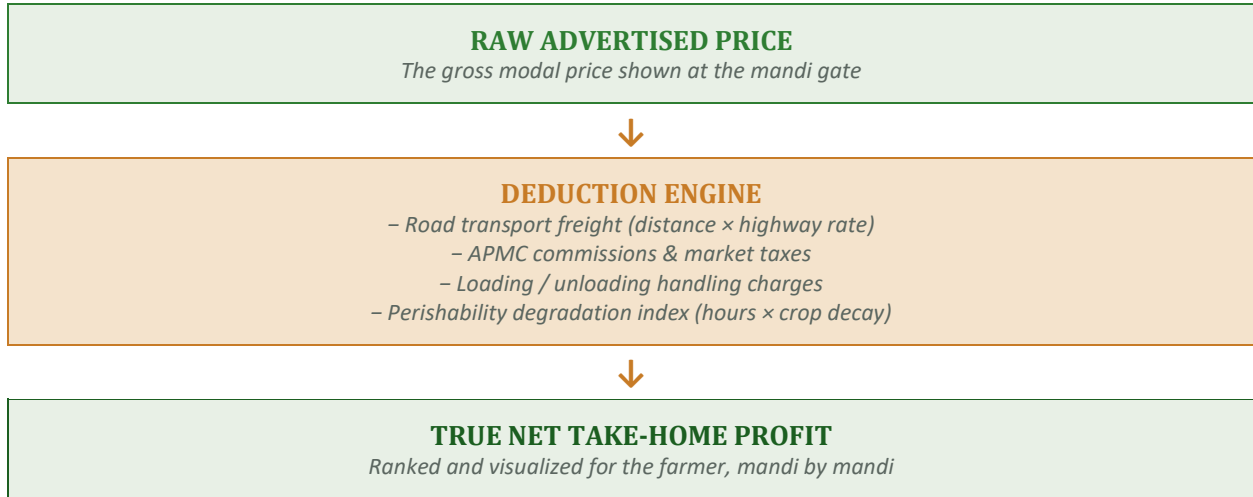
Key Insight

The nearby Kota Mandi delivers ₹3,382 more real profit per quintal than the seemingly superior Delhi price, and over ₹1,700 more per quintal than Mumbai — which results in a total loss once freight and spoilage are accounted for. Gross price alone is a misleading decision signal.

2 The Solution Engineered

To completely eliminate this price illusion, I designed and architected the Smart Mandi Selection & Logistics Platform — a full-stack, data-driven decision engine that calculates real take-home net profit and optimizes agricultural freight for every farmer, in real time.

2.1 How the Engine Thinks: From Raw Price to True Profit



2.2 Core Technological Innovations

1. Five-Factor Net Profit Formulation

Net Profit = Modal Price – (Transport + Handling + Commission + Transit Spoilage)

Spoilage is calculated dynamically as: Modal Price × Perishability Index × (Transit Hours ÷ 24) × 0.15

Scientific decay index by crop — Tomato 0.85, Banana 0.70, Onion 0.25, Potato 0.20, Wheat 0.05.

2. Leaflet GIS Spatial Route Mapping

Colour-coded visual route lines let farmers see why distance matters at a glance:

- **Green** — Maximum profit route
- **Teal** — Moderate profit alternative
- **Amber** — Lower profit / high-distance market

3. Kisan Pool Shared Freight Optimizer

Solves small-batch transport penalties by algorithmically pooling neighbouring farmers' harvests into larger freight vehicles — scaling from a Tata Ace, to an Eicher 14ft, to a 19ft Heavy Truck, up to a 16-Ton Taurus — reducing individual transport costs by 35% to 55%.

4. Zero-App WhatsApp Voice & Text AI Assistant

A multi-lingual conversational bot supporting voice notes in Hindi, Marathi, Gujarati, and English, with instant synthesized voice-note audio replies and GPS location pin support — no app install required.

5. Real-Time Admin Analytics & Dynamic Cost Controls

Live query streaming, price trend curves, and live fee configuration give APMC administrators full control over commission and transport parameters.

3 Step-by-Step User Walkthrough

3.1 Web Application Walkthrough

A. Public Landing Page

smart-mandi-selection.vercel.app

1. View the value proposition — explore the live comparison scenario illustrating real profit differences.
2. Switch languages — click the language selector in the top bar to toggle between English, Hindi (हिंदी), Marathi (मराठी), and Gujarati (ગુજરાતી).
3. Connect to WhatsApp — use the 1-click WhatsApp button to launch the assistant with pre-filled sandbox codes.
4. In-browser Voice AI — click “Try In-Browser Simulator” to test native speech queries directly on your laptop or phone without WhatsApp.

B. Interactive Profit Map (/map)

1. Click “Profit Map” in the top navigation bar.
2. Select your Origin Hub (e.g., Vadodara, Surat, Rajkot, Jaipur, Pune, Nashik, etc.).
3. Select your Crop (Tomato, Onion, Potato, Wheat, Banana) and set your Harvest Quantity via the slider.
4. Read the map routes — green (#1) is maximum net profit, teal (#2) is moderate profit, amber (#3) is lower profit / high distance.
5. Click any mandi pin to view a complete pop-up breakdown of net take-home, gross modal price, total deductions, and travel distance in km.

C. Kisan Pool Shared Freight Optimizer (/pooling)

1. Click “Kisan Pool” in the navigation bar.
2. Use the interactive savings calculator — adjust Solo Quantity (e.g., 12 quintals), Total Pooled Quantity (e.g., 36 quintals), and Distance (e.g., 248 km).

3. The platform dynamically matches the best commercial vehicle and calculates instant savings per quintal.
4. Browse Active Pooling Batches to view live clusters and connect with cluster FPOs in one click.

D. Admin Intelligence Dashboard (/admin/dashboard)

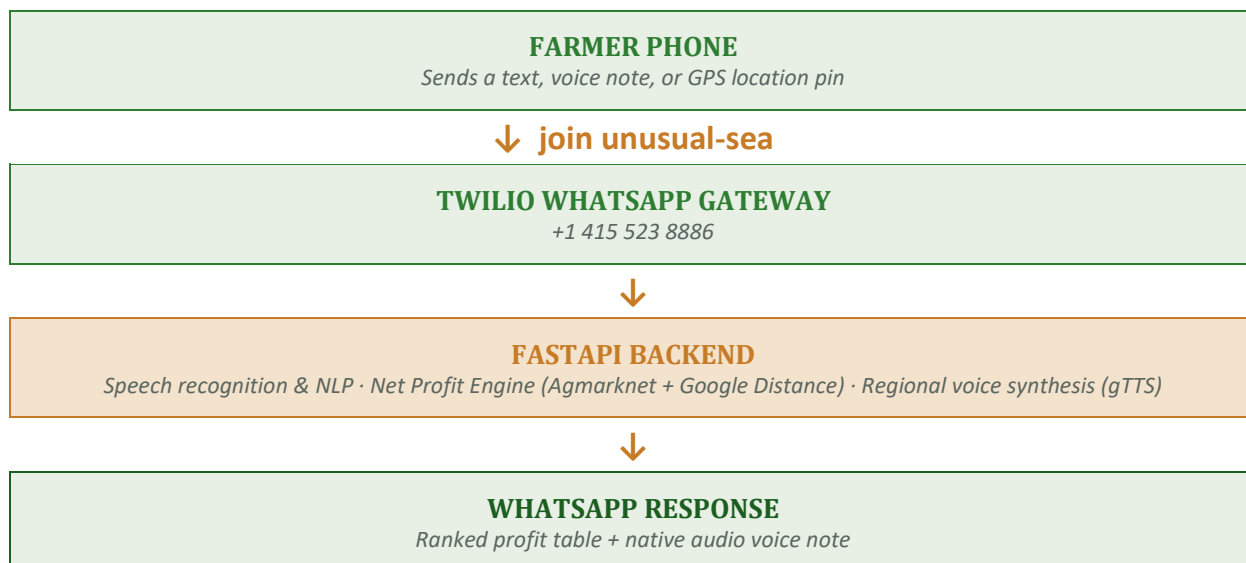
1. Click “Admin Dashboard” or go to /admin/login.
2. Sign in — Username: admin | Password: admin123.
3. View platform KPIs — mandis tracked, active farmers, average savings per quintal, and total queries.
4. Review top crops & mandis — a visual breakdown of most-queried crops and highest-profit mandis.
5. Watch the live farmer query stream — a real-time feed of incoming WhatsApp queries and automated responses.
6. Export data — download complete platform intelligence as a CSV spreadsheet.

E. Price Trends & Cost Parameters (/admin/mandis, /admin/costs)

1. Price Trends — review 30-day historical modal price curves across crops and markets.
2. Cost Parameters — edit mandi-specific commission % (5% to 8%), loading charges, unloading charges, and base transport rates; changes take effect immediately across all calculations.

3.2 WhatsApp Voice & Text AI Assistant Walkthrough

The WhatsApp bot gives farmers zero-app access on the smartphones they already own.



Step 1 — Connect to the Bot Gateway

1. Open WhatsApp and start a chat with +1 (415) 523-8886.
2. Send the activation code: join unusual-sea.
3. Or use the direct 1-click link: wa.me/14155238886?text=join%20unusual-sea
4. You will receive an immediate confirmation that the session is active.

Step 2 — Send Natural Text Queries

Type any natural crop and quantity query, for example:

- “Tomato 20q from Jaipur”
- “कांदा 15 किंटल नाशिक”
- “Potato 30q from Agra”
- “Wheat 50 quintal from Rajkot”

Step 3 — Send Regional Voice Notes (Speech-to-Speech)

Hold the microphone icon in WhatsApp and speak in your native dialect:

- Hindi — “भैया 20 किंटल टमाटर बेचना है जयपुर से”
- Marathi — “नमस्कार, 15 किंटल कांदा नाशिकमधून विकायचा आहे”
- Gujarati — “નમસ્તે, 20 કિવન્ટલ ડુંગળી રાજકોટથી વેચવી છે”

The bot analyzes the voice note and replies with a text summary plus a synthesized voice audio note in your native language, explaining the most profitable mandi.

Step 4 — Share GPS Location Pins

Tap the attachment icon in WhatsApp, select Location, then send your current location. The bot automatically calculates road distances to every nearby APMC mandi without manual typing.

4 Technical Setup & Local Run Guide

4.1 Prerequisites

- Python 3.11+
- Node.js 18+ / 20+

4.2 Run Backend Locally

```
cd backend
.\.venv\Scripts\activate
python -m uvicorn app.main:app --reload --port 8000
```

API Documentation: <http://localhost:8000/docs>

4.3 Run Frontend Locally

```
cd frontend
npm install
npm run dev
```

Web Portal: <http://localhost:5173>

PROJECT LEAD & DEVELOPER

Devansh Rahatal

Created & Architected the Smart Mandi Selection Platform

Repository: [smart-mandi-selection](#)